have targets, download them from the 'Available Packages' window of Android SDK and AVD Manager.

- You can also specify some hardware properties in this dialog box.

Once done, you'll notice a new virtual device created on your AVD and SDK Manager. You can now start this AVD from the 'Start' option in the manager. This will open the emulator in a pop-up.

2.8.1 Emulator or device?

Even though running your application on an emulator is similar to that on a physical device, there are a few exceptions, e.g. visualizing the action of some sensors. This can be tested on a physical device by programming using the USB Debugging Mode.

2.8.2 Controlling the emulator

You interact with the device just like you would with a physical device. You use the mouse to give touch inputs and type on keyboard keys to trigger the simulated device keys. Some keyboard shortcuts:

2.8.3 Limitations

The emulator gives no support for placing or receiving actual phone calls. However, placed and received phone calls can be simulated through the emulator console. It doesn't support USB connection, camera input, headphone jack insert, determination of SD card insert or eject, or Bluetooth.

2.9 Hardware device

Before releasing to users, it's always better to test the application on a real device. You can use any Android device as the environment to run, debug and test the applications you create. SDK tools make installation of the application on the device simple on every compilation.

Set up the device for development by:

1. Declaring the application as Debuggable in the AndroidManifest.xml by adding android:debuggable="true" to the <application> block
2. Turn on USB debugging on the device from Applications > Development
3. Set up your system to detect the device. Windows users need to install drivers for the device from http://developer.android.com/sdk/oem-usb.html. Mac OS X will automatically detect it.

Your device is ready for Debugging, for which you can use the DDMS debugging utility (as described in the User Interface chapter).